

Max Marks: 20

Attempt any two from questions 1, and 2-5 are compulsory



- (i) Find the PCNF for the given well-formed formula:
 $(\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q)$

2

- (ii) Let $S = \{x \in \mathbb{R} | 0 \leq x \leq 1\}$ and $T = \{x \in \mathbb{R} | 0.5 \leq x \leq 5\}$. Find $S - T$ and $S \oplus T$.

2

- (iii) Show that the "greater than or equal" relation is a partial ordering on a set of positive integers.

2

- (iv) If f , g , and h are functions from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = x^2$, $g(x) = x + 5$, and $h(x) = \sqrt{x^2}$, then verify that $fo(goh) = (fog)oh$.

2

2.

What is the maximum height of any AVL tree with 10 nodes? Assume that the height of a tree with a single node is 0. Also construct an AVL tree for the given set of numbers: 8, 7, 1, 2, 6, 5, 3, 4.

4

3.

Differentiate the least upper bound and greatest lower bound. Analyze the time complexity of the given code:

```
A(n)
{
  int i, j, k, n;
  For(i=1; i<=n; i++)
  {
    For(j=1; j<=i^2; j++)
    {
      For(k=1; k<=n/2; k++)
      {
        Printf("SC&SS");
      }
    }
  }
}
```

$$P(1) = 2^4$$

$$Q(1) = 1 + 5 = 6$$

$$h(1) = \sqrt{1} = 1$$

4.

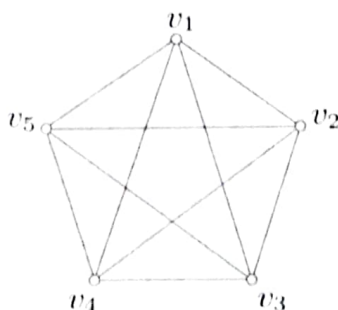
Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

4

5.

Consider a graph $G(V, E)$. Check whether a given graph is bipartite. If it is a bipartite graph then apply the breadth-first search (BFS) using queue data structure otherwise take the counter-example and prove that the bipartite graph is a 2-colorable problem and also show the elements in each layer $L_0, L_1, \dots, L_k$ produced by the BFS starting from the node v_1 , and prove that no edge of G joins two nodes of the same layer.

4



SET-B

Discrete Mathematics (Minor Exam)

Max Marks: 20

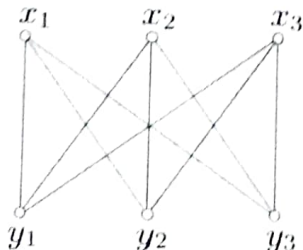
Time: 90 minutes

Attempt any two from questions 1, and 2-5 are compulsory

- 1 (i) Find the PCNF for the given well-formed formula: 2
 $(\neg p \wedge \neg q) \rightarrow (p \leftrightarrow \neg q)$
- (ii) Let $S = \{x \in \mathbb{R} | 0 \leq x \leq 0.6\}$ and $T = \{x \in \mathbb{R} | 0.3 \leq x \leq 1\}$. Find $S - T$ and $S \oplus T$.
- (iii) Show that the "greater than" relation is a partial ordering on a set of positive integers 2
- (iv) If f, g , and h are functions from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = x + 3$, $g(x) = x^2$, and $h(x) = \sqrt{x^2}$, then verify that $fo(goh) = (fog)oh$. 2
2. What is the maximum height of any AVL tree with 12 nodes? Assume that the height of a tree with a single node is 1. 4
 Construct an AVL tree for the given set of numbers: 7,8,2,1,5,6,4,3.
3. Differentiate the least upper bound and greatest lower bound. 4
 Analyze the time complexity of the given code:

```

A(n)
{
  int i, j, k, n;
  For(i=1; i<=n; i++)
  {For(j=1; j<=i^2; j++)
    {For(k=1; k<=n; k++)
      {
        Printf("SC&SS");
      }
    }
  }
}
```
4. 4
 Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."
5. Consider a graph $G(V, E)$, Check whether a given graph is bipartite. If it is a bipartite graph then apply the breadth-first search (BFS) using queue data structure and also show the elements in each layer $L_0, L_1, \dots, L_k$ produced by the BFS starting from the node x_1 , and prove that no edge of G joins two nodes of the same layer. 4



Master of Computer Applications

CS-404: Discrete Mathematics

Max Marks: 50

Max Time: 3 Hrs

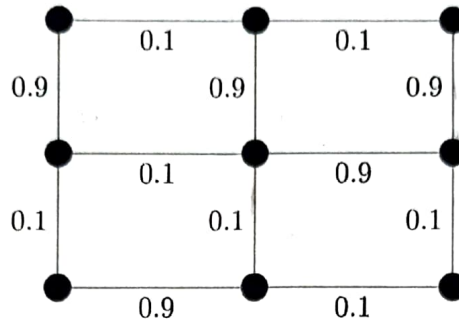
Instructions: Attempt all questions. Parts of the question answered should be together.

- 1 a. In a group of 100 students, 72 can speak English, and 43 can speak Hindi. Based on these data, find the number of students who can speak only English and Hindi and both English and Hindi. 2
- b. Given $A = \{a, b, c, d\}$. Consider the relation R on A :
 $R = \{(a, a), (b, b), (b, c), (c, b), (d, b), (d, d)\}$
Is R Symmetric? Is it transitive? Give reasons. 3
- c. Let x be a real number. Show that $\lfloor 3x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{3} \rfloor + \lfloor x + \frac{2}{3} \rfloor$. 5
- 2 a. A graph contains 21 edges, 3 vertices of degree 4, and all other vertices of degree 2. Find the total number of vertices. 2
- b. Solve the recurrence relation $T(n) = T(n/2) + 1$ for $n > 0$. 3
- c. In a "discrete Mathematics" class with 20 students, the teacher wants to analyze the result at the end of the semester. He/She prepares the following tally of marks obtained by the students: 5

< 40	5
>= 40 and < 50	3
>= 50 and < 60	6
>= 60 and < 70	4
>= 70 and < 80	2

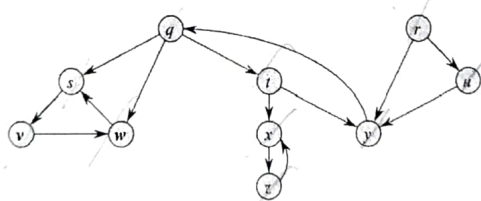
 - (i) While sending the data over the network, the tally is compressed to send the minimum number of bits. Use Huffman's algorithm to generate the optimal binary prefix codes.
 - (ii) With the help of the above table decode the message and find the minimum cost (in terms of bits) for the pattern 10110110000011011.
- 3 a. Differentiate the Prim's and Kruskal's minimum spanning trees with examples. 2
- b. Show that K_6 is a nonplanar graph. 3
- c. Draw a DFA to accept a string of 0's and 1's ending with the string 011. 5

- 4 a. Simplify the expression $(x + y)(x + z)$ using the laws of Boolean algebra. 2
- b. Consider post-order traversal of a binary search tree T produces the following sequence of keys 10, 9, 23, 22, 27, 25, 15, 50, 95, 60, 40, 29. Is it possible to draw a binary search tree for the given post-order? If yes, draw the binary search tree and preorder for the same otherwise provide the reasons. 3
- c. Consider the following undirected graph with edge weight as shown: 5



Find the number of minimum weight spanning trees for the given graph.

- 5 a. What is the Pigeonhole principle? Explain with a suitable example. 2
- b. Run the Depth First Search on the following graph. Show the topological ordering for each vertex, and also consider the edge classification to determine the tree edges, cross edges, forward edges, and back edges. How edge classification helps to determine cycles in directed graphs. 3



- c. Run the Floyd-Warshall algorithm on the given weighted, undirected graph. Show the matrix $D(k)$ that results for each iteration of the outer loop (follow the method as discussed in class). 5

